DEFORMING A TRIPLE SO THAT THE DIVISOR IS CUT OUT BY ONE GLOBAL FUNCTION, WITH FINITE TYPE FLAT RELATIVE DIMENSION ONE MAPS AND INJECTIVE PULLBACK ON CHOW GROUPS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The Gysin homomorphism attached to a triple, which makes the statement meaningful for the deformed triple. Same chapter chow.tex, Definition \ref{definition-gysin-homomorphism}, invoked at proof line 5965. The deformed data $(\mathcal{O}_{X'}, s', i' : D' \to X')$ is again a triple as in the definition, so the pullback $(i')^*$ is defined and the identity in item (4) makes sense.
- Decomposition of a global section of a tensor product of line bundles into two sections. Same chapter chow.tex, label lemma-decompose-section (invoked at proof line 5970). Applied with $\mathcal{L}_1 = \mathcal{O}_X$ and $\mathcal{L}_2 = \mathcal{L}$ to the section s, it produces the commutative diagram together with the finite type, flat of relative dimension one morphisms p and g and the presentation of $D' = D_1$ as the zero scheme of a global section.
- The formulae attached to that decomposition. Same chapter chow.tex, label lemma-decompose-section-formulae (invoked at proof line 5973). Supplies the equality $p^*i^* = (i')^*g^*$ between the maps on Chow groups.
- Injectivity of the pullback along the projection onto the zero scheme in the decomposed situation. Same chapter chow.tex, label lemma-linebundle

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(invoked at proof line 5975). Gives $p^* : \CH_k(D) \to \CH_{k+1}(D')$ injective for all k.

Source reference. Original definition 02T8: sources/stacks-chow/pinned-original-source.tex, source lines 5069–5069. Navigation pointer for \ref{definition-gysin-homomorphism} used inside the counted span (line 5943); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F97: sources/stacks-chow/pinned-original-source.tex, source lines 5822–5822. Navigation pointer for \ref{lemma-decompose-section} used inside the counted span (line 5970); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F98: sources/stacks-chow/pinned-original-source.tex, source lines 5884–5884. Navigation pointer for \ref{lemma-decompose-section-formulae} used inside the counted span (line 5973); the argument that consumes it is retained in the span.

Source reference. Original lemma 0B74: sources/stacks-chow/pinned-original-source.tex, source lines 5737–5737. Navigation pointer for \ref{lemma-linebundle} used inside the counted span (line 5975); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 5941); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 02QL let X be a scheme locally of finite type over S. Let $(\mathcal{L}, s, i : D \to X)$ be a triple as in Definition 02T8. There exists a commutative diagram*

$$\begin{array}{ccc} D' & \xrightarrow{i'} & X' \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle g} \\ D & \xrightarrow{i} & X \end{array}$$

such that

(1) *p and g are of finite type and flat of relative dimension 1,*
(2) *$p^* : \mathrm{CH}_k(D) \to \mathrm{CH}_{k+1}(D')$ is injective for all k,*
(3) *$D' \subset X'$ is the zero scheme of a global section $s' \in \Gamma(X', \mathcal{O}_{X'})$,*
(4) *$p^* i^* = (i')^* g^*$ as maps $\mathrm{CH}_k(X) \to \mathrm{CH}_k(D')$.*

Moreover, these properties remain true after arbitrary base change by morphisms $Y \to X$ which are locally of finite type.

Proof. Observe that $(i')^*$ is defined because we have the triple $(\mathcal{O}_{X'}, s', i' : D' \to X')$ as in Definition 02T8. Thus the statement makes sense.

Set $\mathcal{L}_1 = \mathcal{O}_X$, $\mathcal{L}_2 = \mathcal{L}$ and apply Lemma 0F97 with the section s of $\mathcal{L} = \mathcal{L}_1 \otimes_{\mathcal{O}_X} \mathcal{L}_2$. Take $D' = D_1$. The results now follow from the lemma, from Lemma 0F98 and injectivity by Lemma 0B74. □